

How A.R.T.E.M.I.S Differs From the Closest Published Work

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1 The Two Closest Papers

HULA (Takerngsaksiri et al., ICSE-SEIP 2025, IEEE) deploys a plan-then-approve-then-code agent at Atlassian. Engineers approved the agent's plan in 82 percent of cases, establishing that human approval of an explicit plan is a workable interaction model rather than an obstacle.

SoK: Attack and Defense Landscape of Agentic AI Systems (Kim, Guo and Song, USENIX Security 2026) systematises 51 attacks and 60 defences across 128 papers. It concludes that architectural confinement is more dependable than detecting malicious input, and documents that human approval degrades through decision fatigue.

Together these two represent the state of the art on the two axes our project occupies: the approval loop as an interaction model, and architectural confinement as a security position.

2 What We Take From Them

Our system adopts the plan-and-approve loop that HULA validates, and the confinement-over-detection position that the SoK recommends. Neither is claimed as our contribution.

3 Where We Differ

- **Scope is granted by the user, not assumed.** HULA's agent operates across a repository and the surveyed systems generally assume broad machine access. Our system resolves every path through one component that canonicalises before testing containment, so a file outside a folder the user explicitly granted is unreachable, not merely refused. The SoK argues for this confinement; we apply it to a user-chosen folder on an ordinary desktop.
- **Destruction is unavailable rather than guarded.** There is no approval dialogue for deleting a file because there is no code path to it. A gate the user can click through is a policy; a gate that does not exist is an architecture.
- **Reversal exists.** Neither paper provides a facility to undo an action once executed. We record how to reverse each change before making it, and reverse an approved plan as the single unit the user approved rather than as separate operations.
- **The user is not assumed to be an engineer.** HULA's reviewers are professional engineers; the SoK notes that most defences target organisations with security staff. Our previews are written in plain language for a student or ordinary knowledge worker, and our research with 374 respondents is drawn from that population.
- **Approval is graduated, not uniform.** Actions are classified into four risk tiers, so reads never prompt, reversible changes prompt according to user configuration, and actions that leave the machine always prompt. This directly targets the fatigue the SoK documents.
- **The system discloses what it holds.** A panel reports every file, fact and action retained, and every entry can be deleted. The SoK notes that agent memory is typically an opaque process the user cannot inspect.
- **Beyond code.** HULA addresses software tasks. Our system covers resumption of interrupted work, file organisation, cited synthesis across documents, correspondence and voice, with code assistance as one capability among several and restricted to proposing a difference rather than writing to a file.

4 What We Do Not Claim

We do not solve approval fatigue. The SoK documents it as an open problem in the exact mechanism our design depends upon, and our risk tiers reduce prompt frequency without eliminating the effect. We have written it into our evaluation as something to measure across repeated sessions rather than assert away. We likewise inherit the limitations that a language model can produce unfaithful summaries and unreliable citations; our bounded document set narrows this exposure but does not remove it, and both are reported as known limitations rather than as solved problems.

5 Summary

HULA shows the approval loop works but leaves the agent unbounded and irreversible. The SoK shows confinement is the dependable defence but writes for organisations with security staff. Our project sits in the gap: the validated approval loop, inside a boundary the user grants, with reversal, for one non-expert user on a personal machine.